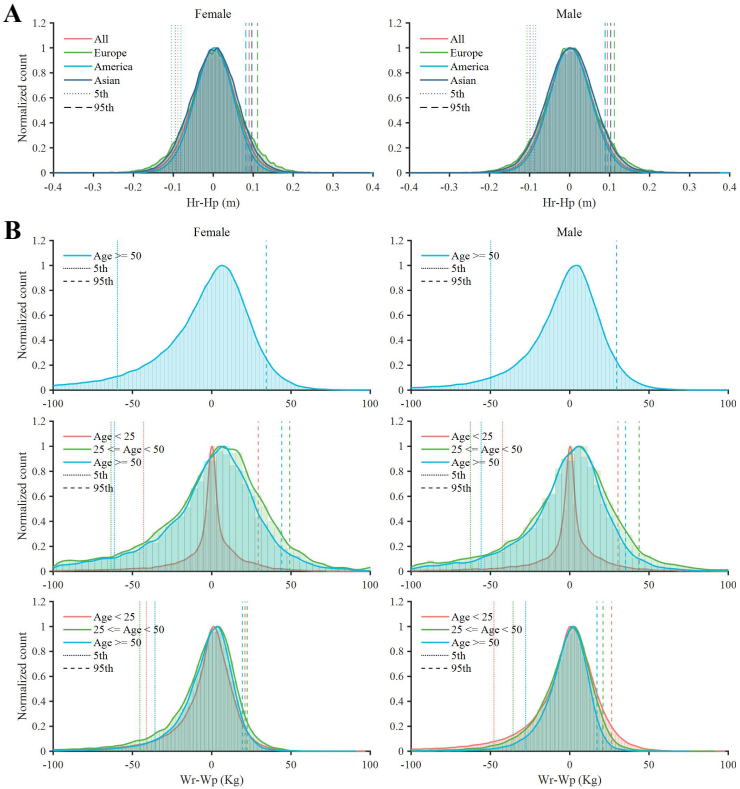

Fig. 5. Distribution curves for weight and height residuals. (A) Comparison of height residual distributions across continents. (B) Comparison of weight residual distributions across continents and age groups. To elucidate age-dependent patterns, the weight residuals were aggregated by continent (top panels: Europe; middle panels: Americas; bottom panels: Asia) and further categorized into three age groups (Age < 25 years, 25 ≤ Age < 50 years, and Age ≥ 50 years). To address the issue of predicted weight approaching infinity as height nears the theoretical upper limit (1/b), an exclusion criterion was applied to omit data points with weight residuals falling outside the biologically plausible range of -100 Kg to 100 Kg. For all distribution curves, the 5th and 95th percentile positions are marked to illustrate symmetry characteristics. The weight and height residuals (Hr-Hp, Wr-Wp) are defined as the differences between the observed values and the constraint-model-predicted values.